

Week 3: Newton's Laws and Forces – Tutorial Sheet

Hope Donglo, Office 24, Department of Physics, University of Ghana

Use the Master Procedure: identify, isolate, draw the free-body diagram, choose axes, predict, write $\sum F = ma$ by components, solve, evaluate. Use $g = 9.80 \text{ m/s}^2$ and treat ropes and pulleys as ideal unless stated. Problem A1 was moved out of the lecture and is solved in full; the topics in section B were moved from Contact 3. Final answers to the rest are at the end.

A. Moved from the lecture

A1. A person in a lift

A 70.0 kg person stands on a scale in a lift. Find the scale reading n when the lift (a) moves upward at constant velocity, (b) accelerates upward at 1.5 m/s^2 , (c) accelerates downward at 1.5 m/s^2 , (d) is in free fall after the cable breaks.

Solution. FBD of the person: $\vec{n}$ up (scale on person), $\vec{w}$ down (Earth on person). Take $+y$ upward: $\sum F_y = n - mg = ma_y$, so $n = m(g + a_y)$. (a) $a_y = 0$: $n = 70.0(9.80) = 686 \text{ N}$, equal to the weight. (b) $a_y = +1.5$: $n = 70.0(11.3) = 791 \text{ N}$, larger than mg . (c) $a_y = -1.5$: $n = 70.0(8.3) = 581 \text{ N}$, smaller than mg . (d) $a_y = -g$: $n = 0$ ("weightless"). *Check:* the scale reading is the normal force, not the weight; the weight mg is the same in all four cases.

B. Practice

B1. Net force

Three forces act on an object: 100 N along $+x$, 60 N along $+y$ and 80 N at 225° from $+x$. Find the magnitude and direction of the net force, and the acceleration of a 5.0 kg object.

B2. Atwood machine

Two blocks of mass 5.0 kg and 3.0 kg hang from the ends of a light rope over a frictionless pulley. Find the acceleration and the tension.

B3. Connected objects with friction

A 4.0 kg block on a table with $\mu_k = 0.25$ is joined by a rope over a pulley to a hanging 2.0 kg block. Find the acceleration and the tension.

B4. Incline: limit of static friction

A block rests on an incline with $\mu_s = 0.60$. What is the largest angle for which it does not slide?

B5. Pulling up an incline

A 20.0 kg crate is pulled at constant speed up a 25° incline by a rope parallel to the slope. The coefficient of kinetic friction is 0.20. Find the tension.

B6. Braking lorry

A box rests on a lorry bed with $\mu_s = 0.40$. What is the greatest deceleration of the lorry for which the box does not slide?

C. Circular dynamics: moved from the lecture

C1. Banked curve

A frictionless banked curve of radius 100 m is to be taken at 20.0 m/s. At what angle must it be banked?

C2. Conical pendulum

A small mass hangs on a 1.00 m string and moves in a horizontal circle with the string at 30° from the vertical. Find its angular speed and the tension for $m = 0.50$ kg.

C3. Vertical circle

A 0.50 kg ball is whirled on a string in a vertical circle of radius 0.80 m. At the top the speed is 4.0 m/s. Find the tension at the top, and the least speed at the top for which the string stays taut.

Answers

- B1 43.5 N at 4.5° above $+x$; $a = 8.7 \text{ m/s}^2$
B2 $a = 2.45 \text{ m/s}^2$; $T = 36.8 \text{ N}$
B3 $a = 1.6 \text{ m/s}^2$; $T = 16.3 \text{ N}$
B4 $\tan \theta = \mu_s$, so $\theta = 31.0^\circ$
B5 $T = mg(\sin \theta + \mu_k \cos \theta) = 118 \text{ N}$
B6 $a = \mu_s g = 3.9 \text{ m/s}^2$
C1 $\tan \theta = v^2/(gR) = 0.408$, so $\theta = 22.2^\circ$
C2 $\omega = \sqrt{g/(L \cos \theta)} = 3.36 \text{ rad/s}$; $T = mg/\cos \theta = 5.7 \text{ N}$
C3 $T = mv^2/R - mg = 5.1 \text{ N}$; $v_{\min} = \sqrt{gR} = 2.8 \text{ m/s}$